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Trade & subcontractor allocation

An independent review of what is built, what is not, and what it would take - read-only, 14 October 2026.

Trade & Subcontractor Allocation – Independent Review

Prepared for: Darren Hopwood - ahead of the CBRE EMEA Technical Directors meeting, 15 October 2026 **Prepared by:** Banksy (Hermes agent), 14 October 2026 **Basis:** read-only review of the current branch (`main`, HEAD `2900dac`) and the live application at instructbrain.com. Supersedes nothing; the typecheck gate was cleared in `2900dac` +1 (test file only, no deploy needed) **Status:** review and recommendations only. No code, data, settings or migrations were changed for this review.

Headline

The brief's eight risks all point at something real. Six are **confirmed** against the code and the live database, two are **partially confirmed** - the guard exists but cannot check a concept that does not yet exist.

The machinery is in better shape than the brief assumes. What is genuinely missing is one layer: **the mapping between a trade and the company appointed to do it.** Everything else - the directory, the per-trade extract, the send log, the frozen published copy, the database-level confidentiality block - is built and working today.

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1. Current-state map

Tables (live `public` schema)

Area	Tables
Directory	`project_directory` (project_id, trade, company_name, is_active, notes), `directory_contacts` (directory_id, name, email, phone, is_primary, receives_copies), `directory_templates`, `directory_template_entries`
Allocation	`findings` (ai_suggested_trade, ai_trade_reasoning, ai_trade_confidence, ai_abstain_reason, assigned_trade, assigned_contact_id, confirmed_by, confirmed_at, is_confidential, lifecycle_state)
Distribution	`distributions`, `trade_access`
Reports	`reports`, `report_versions`, `report_shares`, `photos`, `finding_photos`, `photo_markups`, `audit_log`, `ai_usage_events`
Definitions	`survey_type_definitions`, `report_templates`
Compliance	`compliance_runs`, `compliance_points`, `compliance_entries`, `compliance_actions`
Tenancy	`organisations`, `memberships`, `platform_admins`, `plan_limits`

There is no `subcontractors` table, no `project_trades` table and no `trade_packages` table. A `trade` is stored as text on a directory row, keyed by string. This is the gap the brief identifies, and it is confirmed.

Code and routes

Concern	Where
Grouping and extract documents	`src/lib/distribution.ts` - `groupKeyOf`, `groupForDistribution`, `groupLabelOf`, `selectDistributionFindings`, `buildExtractDocument`, `UNASSIGNED_GROUP`, groupings `trade | area | severity`
Building the send plan	`src/lib/distribution/distribution-data.ts` - matches a directory row by **exact trade string**, picks the `is_primary` contact else the first, falls back to the project fallback recipient
Directory operations	`src/lib/directory/directory-data.ts` - `addDirectoryContact`, `copyDirectoryFromProject`, `importDirectoryGroups` (CSV), `saveDirectoryAsTemplate`, `applyDirectoryTemplate`, `updateFallbackRecipient`
Trade vocabulary	`src/lib/trades/known-trades.ts` - `knownTrades()` derived from the survey definitions; `tradeOptions(existing)` puts the project's own trades first
Bulk trade confirmation	`src/lib/findings/bulk-trade.ts` (`bulkTradeEligible`), `src/lib/review/bulk-accept.ts` (`AcceptPlan`)
Review UI	`src/components/review/trade-organiser.tsx`, `trade-assignment-card.tsx`, `condition-grade-organiser.tsx`
Trade layer switch	`src/components/report/trade-allocation-switch.tsx`; org default `organisations.trade_allocation_enabled`, per-report override `reports.trade_allocation_enabled`
Distribution entry point	`src/components/report/report-actions.tsx` - the **Distribute** button (commit `2900dac`)
Routes	`/reports/$id/distribute`, `/reports/$id/extract/$group`, `/projects/$id/directory`, `/settings/directory`, `/api/public/trade-access/$token`, `/api/public/shared-report(-pdf)/$token`

Database-level guards (verified via `pg_trigger`)

Trigger	Table	What it enforces
`distributions_block_confidential`	distributions	Confidential findings can never enter a subcontractor distribution - enforced in the database, not the UI
`findings_locked_when_issued`	findings	An issued report's findings cannot be edited
`findings_confidentiality_transition`	findings	Guards a change in confidentiality state
`findings_ref_immutable`	findings	Item references cannot be re-pointed
`reports_reference_immutable`, `reports_assign_reference`	reports	References are assigned once and never rewritten
`reports_enforce_allowance`, `photos_enforce_cap`	reports, photos	Plan allowances

Row-level security (verified via `pg_policies`)

Every table in the directory, allocation, distribution and report set carries per-command policies built from `is_org_member()`, `has_org_role(..., ARRAY['owner','admin','surveyor','supervisor'])` and `is_platform_admin()`, resolved through `project_org()`, `report_org()`, `directory_org()` and `directory_template_org()`. Deletes are restricted to owner and admin. Inserts are guarded by `WITH CHECK` on the same role helper - verified, not assumed. **Multi-tenant boundaries are enforced in the database.**

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2. Recent-change review

Six changes landed in the last two days. All were verified on the live application rather than in the test suite alone.

Change	Commit	Verdict
Share links serve the frozen issued PDF, never the live working copy	`ae9cb4c`	Verified live: the PDF door returns `application/pdf` with a `%PDF-` header; drafts return 404
Menus, selects, dialogs white on the working screens	`944e0c2`	Verified live: panel computed to `oklch(1 0 0)` with dark text and a blue-grey edge
One publication rule for every door a report leaves by	`b0e23e5`	Verified live across the report read paths
Health, safety & housekeeping inspection template	`c79c4a1`	Verified live in the template chooser; six sections, four outcomes, no regulatory citation by design
A "short and sweet" report tone	`76c4314`	Verified live in the tone selector: Formal, Easy-going, Sharp, **Short and sweet**, Meticulous, Sarcastic
A distribution entry point	`2900dac`	Verified live by pressing it - it lands on the distribution screen

The gate, and what it missed

Re-run on the current branch, read-only:

Gate	Result
`bun run test`	**942 passed**, 4 skipped (946 across 98 files, 2 skipped) - matches the recorded figure exactly
`bun run lint`	**0 errors**, 246 warnings (all pre-existing `no-explicit-any`) - matches
`bun run typecheck`	**2 errors** - RED

The two typecheck errors are both in `src/lib/__tests__/health-safety-housekeeping.test.ts`, the test file added with the H&S template this afternoon:

- line 21 - `purposeOf(definition.id).length` : `purposeOf` may return null.
- line 49 - `definition.statuses[3].tone` : the index may be undefined.

Two lines, no production code involved - but it means "tests and lint are green" was being read as "the branch is green" when it is not. Testing and lint alone

were the gate that was run; typecheck was not. Worth noting precisely because the brief itself warns that green tests do not prove a working product.

Fragile, and worth knowing: the distribution screen was unreachable by any link in the codebase before `2900dac` - the only way in was to append `/distribute` to a report URL by hand. And the **Directory** item still disappears from the main navigation whenever the trade layer is switched off (`src/components/app-shell.tsx`), so a new account cannot find the address book it needs to fill in in order to switch the layer on. Same class of defect as the missing door, still open.

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3. Risk verification

#	Risk from the brief	Verdict	Evidence
1	Company and trade mapping conflated; duplicated contacts drift	**Confirmed**	`project_directory` holds `trade` and `company_name` on the same row. No company entity exists. One firm across five trades = five rows repeating the same name and contact
2	Multiple subcontractors for one trade ambiguous; first match taken	**Confirmed**	`distribution-data.ts`: `directoryRows.find(candidate => candidate.trade === group.key)` then `contacts.find(is_primary) ?? contacts[0]`. Findings are grouped by trade, so a second firm on the same trade never appears as its own row and nothing warns
3	Case-insensitive in the UI, exact in distribution	**Confirmed**	`groupKeyOf` trims but does not case-fold; the directory lookup is `===`. The picker's dedupe *is* case-insensitive, and `known-trades.ts` comments claim identity is case-insensitive "matching how allocation groups trades" - the comment does not match the grouping code
4	AI uses template trades, not the project's register	**Confirmed**	No project trade register exists. The suggestion vocabulary is `tradesOf(snapshot)` - the report template's own list
5	Free-text trade entry creates drift	**Confirmed, with live evidence**	Stored trades observed in the database included whole sentences, e.g. *"Lifting equipment inspection body (LOLER competent person) with the plant hirer for the MEWP"*. Nothing in a directory can ever match that, so those items fall to the fallback. These are live in the weekly register drafts **FC-LI-001** and **FC-WE-001** in the Demonstration account, so the evidence is checkable today rather than taken on trust
6	Package boundaries and exclusions unmodelled	**Confirmed**	No scope, zone, floor, block or exclusion columns exist on any mapping. `project_directory.notes` is the only free text. Findings carry a location, and distribution can group by area, but that is finding-side, not appointment-side

#	Risk from the brief	Verdict	Evidence
7	Unmapped or ambiguous items need visible triage; the fallback must not look like an answer	**Partially confirmed**	Verified live: the unassigned row is labelled *"No trade assigned"* with *"(fallback recipient)"*, and blocked rows print plain-word reasons ("Groundworks has no contact with an email address in the project directory"). The review screen carries `unsure_trade` and `no_trade_suggested` attention reasons. **Not verified:** whether an extract delivered through the fallback is itself marked as unallocated - that needs a fixture and a send
8	Bulk confirmation needs strict limits	**Partially confirmed**	`bulkTradeEligible` requires no trade assigned, a non-empty suggestion and confidence at or above `BULK_TRADE_CONFIRM_THRESHOLD` (0.80 - the test pins 0.8 eligible, 0.79 not). `bulk-accept.ts` separates what one press may settle from what a person must decide, and ties its `blocked` flag to the publish gate. **But** it cannot check "trade is in the project register" or "package is unambiguous", because neither concept exists

The brief's checklist of "existing concepts observed" is accurate in every particular: the template supplies the trade list; AI suggestion and human confirmation are stored in separate columns (`ai_suggested_trade`, `assigned_trade`); directory rows carry trade, company, active state and a primary contact; distribution groups by assigned trade and uses the primary contact; a project-level fallback exists; confidential findings are blocked at database level; and the one-press confirmation uses a real 80% threshold.

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4. Recommended user journey

1. **Project setup - one screen.** Tick the trades this job actually has, from a standard library. Project-specific trades can be added deliberately, never typed by accident. Mark one firm as covering Groundworks, Drainage, Reinforced concrete and Below-ground utilities, and the project knows the structure rather than a list of labels.
2. **Add each subcontractor once.** One company record, many trades. Save the whole thing as a template so the next project starts in one click.

3. **AI suggests, from the project's list only.** Confidence and reasoning shown; it abstains when uncertain, and it never names a company as a fact.
4. **A person confirms the trade**, and where two packages could apply, chooses the package. Ambiguity is never resolved by taking the first match.
5. **Pre-flight before anything sends** - unassigned items, missing mappings, ambiguous packages, missing or invalid contacts, excluded scope, unconfirmed findings, confidential findings.
6. **One extract per subcontractor**, covering every trade that firm holds, with each line still showing its trade so their own crews know. Not five emails to the same firm.
7. **Send deliberately.** A person presses the button. Every send is recorded against the report and the issued version.
8. **Issued reports stay frozen.** Changing an appointment afterwards never rewrites an issued report or the PDF an existing share link serves.

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5. Data model proposal – smallest safe change

Keep everything that works. Add the missing layer beside it.

New tables

- `subcontractors` - organisation-scoped company record: id, organisation_id, name, notes, is_active.
- `project_trades` - the project's active register: project_id, trade_id, label, is_active, sort_order.
- `trade_packages` - the mapping: project_id, trade_id, subcontractor_id, zone_or_floor, scope_note, exclusion_note, is_active, effective_from, effective_to.
- `trade_library` - canonical trades: id, label, aliases. Seed it from the trades the survey definitions already carry in `survey_type_definitions` rather than inventing a new vocabulary.
- `findings.confirmed_package_id` - nullable foreign key. Keep `assigned_trade` exactly as it is.

Migration and backfill

1. Derive `project_trades` from the distinct trades already present per project (findings plus directory rows).
2. Derive `subcontractors` from distinct `company_name` within each organisation.

3. Derive `trade_packages` from each existing `project_directory` row - one package per row.
4. Leave `project_directory` in place and keep it working; move the UI to the new tables first, then retire the old rows. No downtime, no rewrite of any issued report or version document.

Security

New tables take the same policy pattern as the existing ones: `is_org_member()` for reads, `has_org_role(..., ARRAY['owner','admin','surveyor','supervisor'])` for writes, owner/admin for deletes, `WITH CHECK` on insert, and a `project_org()`-style resolver. Verified as the house pattern across fourteen existing tables.

Compatibility

Issued reports, version documents and existing share links are untouched by any of this. The `findings_locked_when_issued` trigger already prevents post-issue edits; mapping changes are project-level and must never reach a version document.

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6. Acceptance criteria and tests

#	Scenario	Must be true
1	One subcontractor, four trades	One extract to that firm containing all four trades' items, each line stamped with its trade
2	Two subcontractors, one trade, different floor ranges	Two package rows; items resolve by zone; an item matching neither zone is flagged, never silently assigned
3	Valid trade, no mapped company	Pre-flight blocks the row with a plain-word reason
4	Null or low-confidence suggestion	Not bulk-eligible; appears as an attention item
5	Suggestion not in the project register	Not bulk-eligible; surfaced, not written
6	Case and spacing variations	Both resolve to one canonical trade id; one package, one extract
7	Exclusion that appears to match	Flagged ambiguous; not sent
8	Missing recipient email	Row blocked before sending (already true live)
9	Confidential finding involving a person	Absent from every extract - asserted at the database level, not the UI
10	Report reopened while a share link is live	The link still serves the frozen issued PDF (verified live today)
11	Mapping changed after issue	Issued version and audit trail byte-identical
12	High-confidence suggestion, ambiguous package	Not bulk-confirmed
13	Pre-flight	Lists every unresolved mapping before any send
14	New tables	RLS tests per role: member reads, surveyor writes, admin deletes, another organisation sees nothing

7. Prioritised plan

First, before anything else

0. Clear the two typecheck errors on `main`, and make `test + lint + typecheck` the gate every future change is measured against. Two lines; it should not sit red.

Must fix before this is promised to a client

1. Package model with boundaries and exclusions - risks 1, 2 and 6. This is the whole point of the brief.
2. Project trade register, closed AI vocabulary and stable trade ids - risks 3, 4 and 5.
3. One extract per subcontractor rather than per trade.
4. Pre-flight completeness, and bulk confirmation that refuses an ambiguous package or an unregistered trade - risks 7 and 8.
5. The fallback labelled as triage, never as an answer.

Next, once the above is settled

6. Directory and package setup UI, with the template/CSV path moved onto the new model.
7. Effective dating on packages, so a change of appointment is recorded rather than overwritten.
8. The **Directory** navigation item shown whenever the trade layer is on, with a line explaining it when it is off.

Not doing, deliberately - per the brief, no trade scoring, no subcontractor league tables, no performance rankings. The tool documents and distributes observations; it does not rate companies.

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8. Meeting-ready summary

What is already true today, and can be shown

- A project can hold its trades, with a company and a named contact against each, and that setup can be saved as a template so the next project starts in one click.
- Each item can be attributed to a trade, with the AI's suggestion, its confidence and its reasoning shown, and a person confirming before

anything is treated as final.
- Each trade receives **only its own items** - verified live: the extract carries the line *"This extract lists only the items attributed to Groundworks on this report. It is not the full report."*
- Nothing sends automatically. Every send is one deliberate press and is recorded.
- Published reports and their links serve a frozen copy. Reopening a report does not disturb a link.
- Findings involving a person cannot reach a subcontractor - blocked in the database, not just hidden in the interface.
- Menus, screens and the published PDF are one coherent piece of software, not a prototype.

What is proposed, and would be built

- A project-level map of **trades to subcontractors**: twenty trades, five firms, one firm covering groundworks, drainage, civils and concrete - set up once, reused on every project.
- The AI choosing only from that approved register, never inventing a trade that no firm covers.
- Where one trade is split between firms by floor, zone or block, the split is recorded and a person resolves it - ambiguity is shown, never silently guessed.
- **One extract per firm**, not one per trade.
- A pre-flight check before any send, so missing mappings and missing contacts are visible before anything leaves the building.

Two honest limits for the room

- Where a project has no mapping yet, the tool will say so. It does not know who is responsible, and it will not guess.
- Nothing in this rates a subcontractor. It records what was observed and who was appointed.

Questions worth asking CBRE - the brief lists the right ones: whether responsibility is managed by trade, package, or zone; whether one firm covers several trades and one trade splits across firms; where scope boundaries and exclusions are recorded; who may confirm responsibility; what should happen when an issue cannot be allocated confidently; and what audit evidence is required when an issue is allocated, changed or distributed.

Prepared read-only. No code, data, settings or migrations were changed for this review. Running tests, lint and type checks was the only execution performed.

Prepared by Banksy (Hermes agent) for Darren Hopwood, 14 October 2026. Read-only review: no code, data, settings or migrations were changed. Re-verified against `main` at `2900dac`.